

# **Operational Research Operations Research (OR)**

**Aim at **optimization** of a variety of operations or activities in corporate, social, or engineering systems using mainly **mathematical** models**

**Support quantitatively for decision makers' **most reasonable** decision**

### **Characteristics of OR**

- (1) Model approach**
- (2) Look for most desirable (preferable) solution**
- (3) Interdisciplinary approach**

**Optimization of objective function under some constraint**

- A. Discovery of problems**
- B. Collection of information**
- C. Mathematical modeling of problems**
- D. Solution of mathematical model**
- E. Testing the solution**
- F. Execution of problem solving and feedback of results**

### **Necessary expertise for learning Operational Research (OR)**

**Basic calculus**

**Linear algebra**

**Probability & statistic**

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**1.Linear programming(1)**

## Problem 1

A company manufactures and sells Products A and B. When manufacturing A, the necessary numbers of part P1 and part P2 are one and three, respectively. When manufacturing B, the necessary numbers of part P1 and part P2 are two and two, respectively. The maximum supplies of parts P1 and P2 are 40 and 60, respectively. The profits of one product A and one product B are \$2,500 and \$4,000, respectively. How many products A and B must be manufactured to maximize the profit?

### Mathematical formulation of Problem 1

The number of Product A and Product B manufactured are  $x$  and  $y$ , respectively.

Constraints:

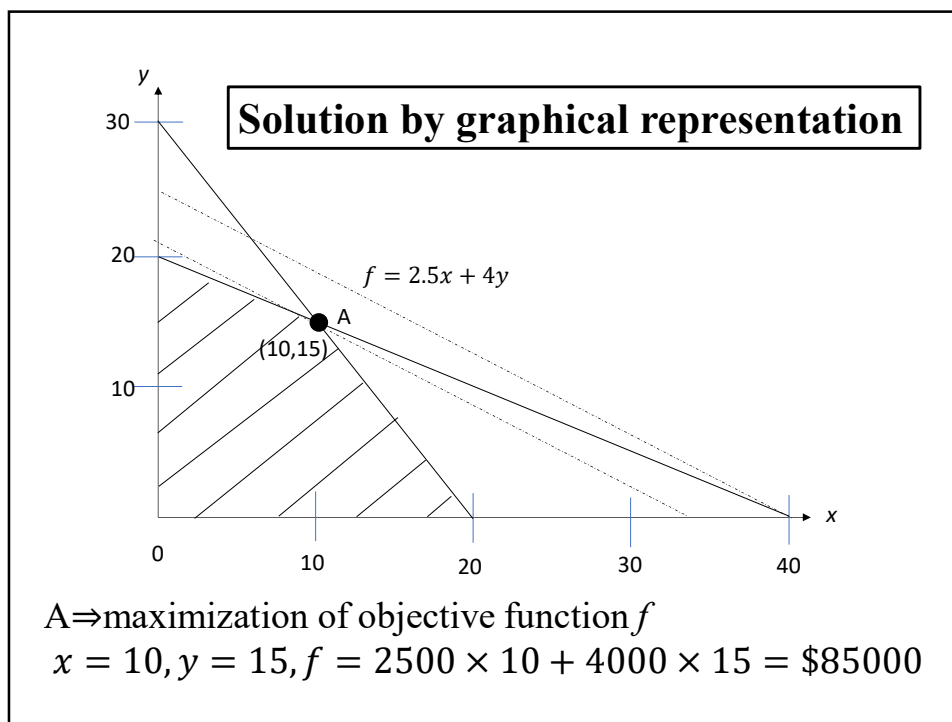
$$\text{P1: } x + 2y \leq 40 \quad (1)$$

$$\text{P2: } 3x + 2y \leq 60 \quad (2)$$

$$x, y \geq 0 \quad (3)$$

Objective function:

$$f(x, y) = 2500x + 4000y \quad (4) \Rightarrow \text{maximization}$$



## Problem 2

A company is faced with a problem to advertise its product to Magazines M1 and M2. The company has a budget of \$2,000. The cost of advertisement per one page in Magazine M1 and M2 are \$200 and \$400, respectively. The numbers of potential customers of M1 and M2 are 1500/page and 600/page, respectively. This company wants to suppress the number of potential customers to equal or less than 6,000. The advertisement effects of M1 and M2 are 7,000 and 5,000, respectively. The number of advertisement pages must be equal or less than 6 pages. How many pages of advertisement to M1 and M2 should be done to maximize the advertisement effect?

## Answer

$$x=2.5, y=3.75, f=36.25 \times 1000=36,250$$

## Problem 3

A company manufactures Products A and B. When manufacturing a one kilogram of A, the necessary material, the necessary electricity and the necessary labor are 9 tons, 4 kWh and 3 workers per day, respectively. When manufacturing a one kilogram of B, the necessary material, the necessary electricity and the necessary labor are 4 tons, 5 kWh and 10 workers per day, respectively. The available material, electricity and labor are 360 tons, 200 kWh, and 300 workers per day, respectively. Products A and B per one kilogram give rise to profits of \$70,000 and \$120,000, respectively. How many products A and B must be manufactured to maximize the profit?

**Answer**

$$x=20, y=24, f=7 \times 20 + 12 \times 24 = 428$$

**2.Linear programming(2)**

### Problem 1(Repost: P1 of L1)

A company manufactures and sells Products A and B. When manufacturing A, the necessary numbers of part P1 and part P2 are one and three, respectively. When manufacturing B, the necessary numbers of part P1 and part P2 are two and two, respectively. The maximum supplies of parts P1 and P2 are 40 and 60, respectively. The profits of one product A and one product B are \$2,500 and \$4,000, respectively. How many products A and B must be manufactured to maximize the profit?

#### Solution by simplex tableau

Constraints:

$$\text{P1: } x + 2y + \lambda_1 = 40 \quad (1)$$

$$\text{P2: } 3x + 2y + \lambda_2 = 60 \quad (2)$$

$$x, y, \lambda_1, \lambda_2 \geq 0 \quad (3)$$

Objective function:

$$f(x, y) = 2500x + 4000y \quad (4)$$

$\Rightarrow$ maximization



### simplex tableau

|        |       | $v_j$       | 0  | 2.5  | 4  | 0           | 0           |               |            |
|--------|-------|-------------|----|------|----|-------------|-------------|---------------|------------|
|        | $v_i$ | variable    | S  | x    | y  | $\lambda_1$ | $\lambda_2$ | $\theta_i$    | Row number |
| Step 1 | 0     | $\lambda_1$ | 40 | 1    | 2  | 1           | 0           | $40/2 = 20$   | (1-1)      |
|        | 0     | $\lambda_2$ | 60 | 3    | 2  | 0           | 1           | $60/2 = 30$   | (1-2)      |
|        |       | $z_j - v_j$ | 0  | -2.5 | -4 | 0           | 0           |               | (1-3)      |
| Step 2 | 4     | y           | 20 | 0.5  | 1  | 0.5         | 0           | $20/0.5 = 40$ | (2-1)      |
|        | 0     | $\lambda_2$ | 20 | 2    | 0  | -1          | 1           | $20/2 = 10$   | (2-2)      |
|        |       | $z_j - v_j$ | 80 | -0.5 | 0  | 2           | 0           |               | (2-3)      |
| Step 3 | 4     | y           | 15 | 0    | 1  | 0.75        | -0.25       |               | (3-1)      |
|        | 2.5   | x           | 10 | 1    | 0  | -0.5        | 0.5         |               | (3-2)      |
|        |       | $z_j - v_j$ | 85 | 0    | 0  | 1.75        | 0.25        |               | (3-3)      |

| Step 2      |    |      |    |             |             |       |
|-------------|----|------|----|-------------|-------------|-------|
|             | S  | x    | y  | $\lambda_1$ | $\lambda_2$ |       |
| (1-1)/2     | 20 | 0.5  | 1  | 0.5         | 0           | (2-1) |
| (1-2)       | 60 | 3    | 2  | 0           | 1           |       |
| -(2-1)X2    | 40 | 1    | 2  | 1           | 0           |       |
|             | 20 | 2    | 0  | -1          | 1           | (2-2) |
| (1-3)       | 0  | -2.5 | -4 | 0           | 0           |       |
| -(2-1)X(-4) | 80 | 2    | 4  | 2           | 0           |       |
|             | 80 | -0.5 | 0  | 2           | 0           | (2-3) |

| Step 3                 |    |      |   |             |             |       |
|------------------------|----|------|---|-------------|-------------|-------|
|                        | S  | x    | y | $\lambda 1$ | $\lambda 2$ |       |
| $(2-2)/2$              | 10 | 1    | 0 | -0.5        | 0.5         | (3-2) |
| $(2-1)$                | 20 | 0.5  | 1 | 0.5         | 0           |       |
| $-(3-2) \times 0.5$    | 5  | 0.5  | 0 | -0.25       | 0.25        |       |
|                        | 15 | 0    | 1 | 0.75        | -0.25       | (3-1) |
| $(2-3)$                | 80 | -0.5 | 0 | 2           | 0           |       |
| $-(3-2) \times (-0.5)$ | 5  | 0.5  | 0 | -0.25       | 0.25        |       |
|                        | 85 | 0    | 0 | 1.75        | 0.25        | (3-3) |

## Answer

$$x=15, y=10, f=2500 \times 10 + 4000 \times 15=85,000$$

## Problem 2 (Repost: P3 of L1)

A company manufactures Products A and B. When manufacturing a one kilogram of A, the necessary material, the necessary electricity and the necessary labor are 9 tons, 4 kWh and 3 workers per day, respectively. When manufacturing a one kilogram of B, the necessary material, the necessary electricity and the necessary labor are 4 tons, 5 kWh and 10 workers per day, respectively. The available material, electricity and labor are 360 tons, 200 kWh, and 300 workers per day, respectively. Products A and B per one kilogram give rise to profits of \$70,000 and \$120,000, respectively. How many products A and B must be manufactured to maximize the profit?

**Answer**

$$x=20, y=24, f=7 \times 20 + 12 \times 24 = 428$$

**3.Nonlinear programming(1)**

## Problem 1

We transport the quantity  $x_{ij}$  from factory  $F_i$  ( $i=1,2$ ) to warehouse  $W_j$  ( $j=1,2,3$ ). The transportation cost is given by

$$g(x_{ij}) = 0.2x_{ij} + 0.01x_{ij}^2$$

Calculate the quantity  $x_{ij}$  so that total transportation cost  $TC$  is minimized. The shipping amount of each factory and the expected demand of each warehouse are given by the table below.

|                 | W1       | W2       | W3       | shipping amount |
|-----------------|----------|----------|----------|-----------------|
| F1              | $x_{11}$ | $x_{12}$ | $x_{13}$ | 31              |
| F2              | $x_{21}$ | $x_{22}$ | $x_{23}$ | 19              |
| Expected demand | 10       | 15       | 25       | 50              |

$$\text{Transportation cost } TC = \sum_{i=1}^2 \sum_{j=1}^3 (0.2x_{ij} + 0.01x_{ij}^2)$$

$$\sum_{i=1}^2 \sum_{j=1}^3 (0.01x_{ij}^2) \Rightarrow \text{Minimization}$$



$$x_{11} = x, x_{12} = y, x_{13} = z$$

$$x + y + z = 31$$

$$x^2 + y^2 + z^2 + (10 - x)^2 + (15 - y)^2 + (25 - z)^2$$

$$\Rightarrow \text{Minimization}$$

### Lagrange multipliers method

$$g(x, y) = 0$$

Maximization of  $f(x, y)$

$$L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$$

If  $(\alpha, \beta)$  is an extreme value,  
 $(\alpha, \beta)$  is a solution of

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial y} = \frac{\partial L}{\partial \lambda} = 0$$

### Example

$$x^2 + y^2 = 1$$

Obtain the maximum value of  $f(x, y) = 2x + 3y$

$$L(x, y, \lambda) = 2x + 3y - \lambda(x^2 + y^2 - 1)$$

$$\frac{\partial L}{\partial x} = 2 - 2x\lambda = 0$$

$$\frac{\partial L}{\partial y} = 3 - 2y\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = -(x^2 + y^2 - 1) = 0$$

$$x = \frac{2}{3}y$$

$$(x, y) = \left(\pm \frac{2}{\sqrt{13}}, \pm \frac{3}{\sqrt{13}}\right)$$

$$(x, y) = \left(\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right) \Rightarrow \text{maximum } \sqrt{13}$$

$$(x, y) = \left(-\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}\right) \Rightarrow \text{minimum } -\sqrt{13}$$

$$Q = x^2 + y^2 + z^2 + (10 - x)^2 + (15 - y)^2 + (25 - z)^2 - \lambda(x + y + z - 31)$$

$$\begin{aligned} \frac{\partial Q}{\partial x} &= 2x - 2(10 - x) - \lambda = 0, \\ 4x - 20 - \lambda &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial Q}{\partial y} &= 2y - 2(15 - y) - \lambda = 0, \\ 4y - 30 - \lambda &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial Q}{\partial z} &= 2z - 2(25 - z) - \lambda = 0, \\ 4z - 50 - \lambda &= 0 \end{aligned}$$

$$2x - 10 - \frac{\lambda}{2} = 0$$

$$2y - 15 - \frac{\lambda}{2} = 0$$

$$2z - 25 - \frac{\lambda}{2} = 0$$

$$2(x + y + z) - 50 - \frac{3}{2}\lambda = 0$$

$$12 - \frac{3}{2}\lambda = 0 \quad \therefore \lambda = 8, x = 7, y = \frac{19}{2}, z = \frac{29}{2}$$

$$\therefore x_{11} = 7, x_{12} = \frac{19}{2}, x_{13} = \frac{29}{2}, x_{21} = 3, x_{22} = \frac{11}{2}, x_{23} = \frac{21}{2}$$

**Problem 2**

Company A advertises its new product in two newspapers P and Q. The advertisement cost of P and Q are ten thousands dollars. The budgeted for advertisement is eighty thousands dollars. The total number of advertisement is eight times. Company A advertises  $x$  times in newspaper P and  $y$  times in newspaper Q. For simplicity, we assume that  $x$  and  $y$  are not necessarily integer numbers. The effects of advertisement  $R$  is given by  $R = 80x - 4x^2 + 100y - 5y^2 + 15$  (ten thousands people).

Obtain  $x$  and  $y$  that maximizes  $R$ .

**Answer**

$$x = \frac{10}{3} = 3.3, y = \frac{14}{3} = 4.7$$

## 4. Nonlinear programming(2)

### Problem 1

One company plans to advertise its product four times on newspapers P, Q, and R. The advertisement effect of each newspaper as a function of frequency is given by the table below. Obtain the optimum advertisement strategy that maximizes the advertisement effect.

|           | P          | Q          | R          |
|-----------|------------|------------|------------|
| effect    | $f_1(x_1)$ | $f_2(x_2)$ | $f_3(x_3)$ |
| frequency |            |            |            |
| 0         | 0          | 0          | 0          |
| 1         | 3          | 1          | 6          |
| 2         | 5          | 3          | 7          |
| 3         | 6          | 8          | 8          |
| 4         | 7          | 10         | 9          |



|             | 1 | 2 | 3 |                           |
|-------------|---|---|---|---------------------------|
| 1 magazine  | 4 |   | 0 | 3 cases; P(4), Q(4), R(4) |
| 2 magazines | 3 | 1 | 0 | 6 cass: 3C1X2             |
| 2 magazines | 2 | 2 | 0 | 3 cases: 3C1              |
| 3 magazines | 2 | 1 | 1 | 3 cases: 3C1              |
|             |   |   |   | 15 cases                  |

|    |       | P |    |    |   |   |   |
|----|-------|---|----|----|---|---|---|
| Q  | x1    |   | 0  | 1  | 2 | 3 | 4 |
|    | f(x1) |   | 0  | 3  | 5 | 6 | 7 |
| x2 | f(x2) |   |    |    |   |   |   |
| 0  | 0     |   | 0  | 3  | 5 | 6 | 7 |
| 1  | 1     |   | 1  | 4  | 6 | 7 |   |
| 2  | 3     |   | 3  | 6  | 8 |   |   |
| 3  | 8     |   | 8  | 11 |   |   |   |
| 4  | 10    |   | 10 |    |   |   |   |

P,Q:3  
P:0 Q:3

P,Q,R:4  
P,Q:3 R:1

|     | P, Q    |   |    |    |    |    |
|-----|---------|---|----|----|----|----|
| R   | x       | 0 | 1  | 2  | 3  | 4  |
|     | F2(x)   | 0 | 3  | 5  | 8  | 11 |
| c-x | f3(c-x) |   |    |    |    |    |
| 0   | 0       | 0 | 3  | 5  | 8  | 11 |
| 1   | 6       | 6 | 9  | 11 | 14 |    |
| 2   | 7       | 7 | 10 | 12 |    |    |
| 3   | 8       | 8 | 11 |    |    |    |
| 4   | 9       | 9 |    |    |    |    |

Optimum solution    P:0 Q:3 R:1

|         |   |   |    |   |    |    |    |    |    |    |    |    |    |    |    |
|---------|---|---|----|---|----|----|----|----|----|----|----|----|----|----|----|
|         | 1 | 2 | 3  | 4 | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| Magzine |   |   |    |   |    |    |    |    |    |    |    |    |    |    |    |
| P       | 4 | 3 | 3  | 2 | 2  | 2  | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0  |
| Q       | 0 | 1 | 0  | 2 | 1  | 0  | 3  | 2  | 1  | 0  | 4  | 3  | 2  | 1  | 0  |
| R       | 0 | 0 | 1  | 0 | 1  | 2  | 0  | 1  | 2  | 3  | 0  | 1  | 2  | 3  | 4  |
| Effect  | 7 | 7 | 12 | 8 | 12 | 11 | 11 | 12 | 11 | 11 | 10 | 14 | 10 | 9  | 9  |

Problem 2

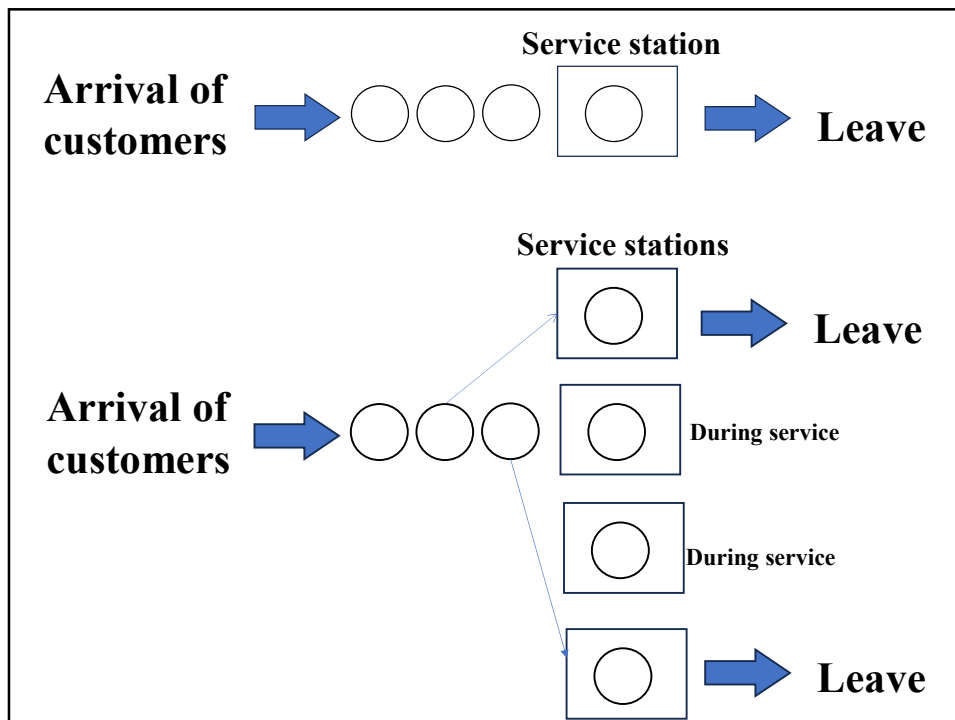
One company plans to advertise its product six times on newspapers A, B, C, D, and E. The advertisement effect of each newspaper as a function of frequency is given by the table below. Obtain the optimum advertisement strategy that maximizes the advertisement effect.

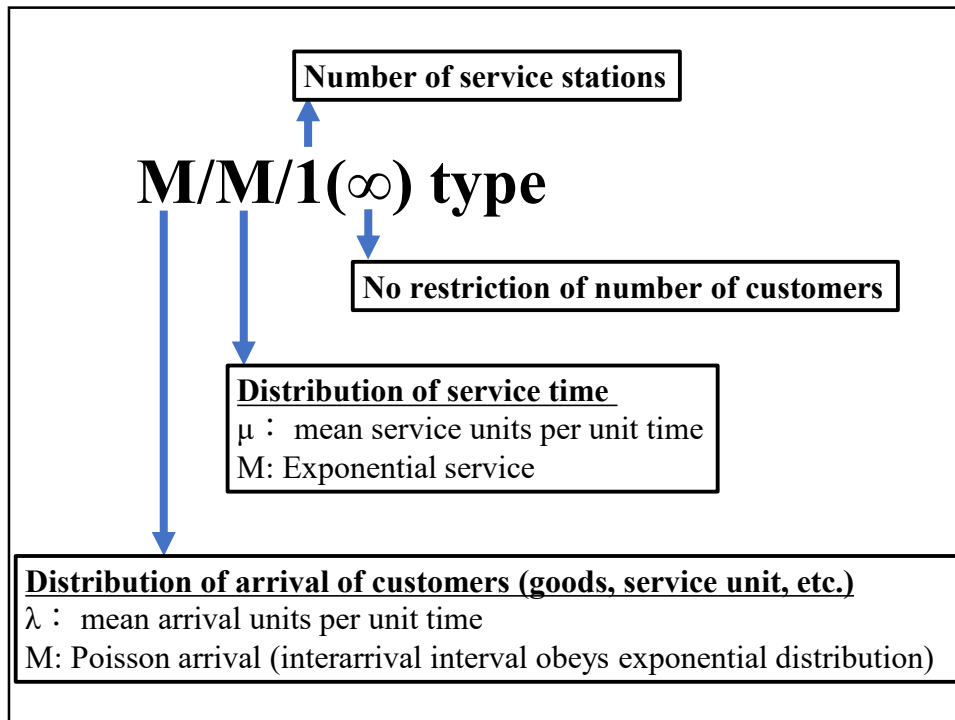
|           | A          | B          | C          | D          | E          |
|-----------|------------|------------|------------|------------|------------|
| effect    | $f_1(x_1)$ | $f_2(x_2)$ | $f_3(x_3)$ | $f_4(x_4)$ | $f_5(x_5)$ |
| frequency |            |            |            |            |            |
| 0         | 0          | 0          | 0          | 0          | 0          |
| 1         | 6          | 2          | 10         | 4          | 8          |
| 2         | 10         | 8          | 10         | 10         | 12         |
| 3         | 11         | 12         | 10         | 12         | 12         |
| 4         | 12         | 14         | 10         | 14         | 10         |
| 5         | 13         | 15         | 10         | 16         | 10         |
| 6         | 14         | 15         | 10         | 13         | 8          |

Optimum solution

|   | A | B | C | D | E | Sum |
|---|---|---|---|---|---|-----|
| ○ | 2 | 0 | 1 | 2 | 1 | 6   |
| △ | 1 | 0 | 1 | 2 | 2 | 6   |

## 5. Queuing problem





## M/M/1(∞) type

Mean arrival rate  $\lambda$  (arrival units/time)  
 Mean service rate  $\mu$  (service units/time)  
 Load factor or traffic intensity  $\rho = \lambda / \mu < 1$

## Basic formulas

**(1) Probability ( $P_n$ ) of  $n$  service units in the system (Both waiting lines and a service station are empty)**

$$P_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right) = \rho^n (1 - \rho), \rho < 1, n \geq 0$$

**(2) Mean number of units in the system (both waiting units and a unit receiving service)**

$$E(n) = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho}, \rho < 1, n \geq 0$$

**(3) Mean number of waiting units in the system**

$$E(m) = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{\rho^2}{1 - \rho}, \rho < 1, n \geq 0$$

**(4) Mean number of waiting units in the system when there was actually a waiting unit**

$$E(m|m > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \rho}, \rho < 1, n \geq 0$$

**(5) Mean time spending in the system when a unit actually arrives at the system**

$$E(v) = \frac{1}{\mu - \lambda}, \rho < 1, n \geq 0$$

**(6) Mean waiting time of an arrived unit**

$$E(w) = \frac{\lambda}{\mu(\mu - \lambda)}, \rho < 1, n \geq 0$$

**(7) Mean waiting time of an arrived unit that actually waited for the service**

$$E(w|w > 0) = \frac{1}{\mu - \lambda}, \rho < 1, n \geq 0$$

## Problem 1

We assume that customers arrive at a customer service helpdesk of company A. As a result of observing the distributions of arrival and service, we found that the arrival distribution obeys Poisson distribution with the mean arrival interval of 5 minutes. The distribution of service time obeyed exponential distribution with the mean service time of 4 minutes.

- (1) Calculate the probability that a customer must wait at the service helpdesk.
- (2) Calculate the number of waiting persons when a customer must wait.
- (3) Calculate the mean waiting time of a customer.

Mean arrival rate  $\lambda = \frac{1}{5} = 0.2$  persons/min

Mean service rate  $\mu = \frac{1}{4} = 0.25$  persons/min

$$\rho = \frac{\lambda}{\mu} = \frac{0.2}{0.25} = 0.8 < 1$$

$$(1) P(n \geq 1) = 1 - P_{(n=0)} = 1 - \rho^0(1 - \rho) = 0.8$$

$$(2) \frac{\mu}{\mu - \lambda} = \frac{0.25}{0.25 - 0.2} = 5 \text{ persons}$$

$$(3) \frac{\lambda}{\mu(\mu - \lambda)} = \frac{0.2}{0.25(0.25 - 0.2)} = 16 \text{ min}$$

## Problem 2

There is a repair shop with one repairman. As a result of observing the distributions of arrival and service, we found that the arrival distribution obeys Poisson distribution with the mean arrival rate of 10 persons per day. The distribution of service time obeyed exponential distribution with the mean repair time of 30 minutes.

(1) Assume that one day consists of 8 working hours.

Calculate the time when the repairman is not engaged in a repair service out of eight working hours.

(2). When a customer arrives at the service shop, calculate how many repair jobs the repairman has. It is assumed that one customer brings one product for repair to the repair shop.

## Answer

$$(1) P_0 = \frac{3}{8}$$

Time of no engagement = 3 hours

$$(2) E(n) = \frac{5}{3} \text{ repair jobs}$$



### Problem 3

We assume that the mean arrival rate in M/M/1 queuing system is 2 persons/hour. We would like to make the following probability  $P_3$  equal or less than 0.125.

$P_3$ : the probability that the number of persons in a system (a person who is undergoing a service and a person who is waiting for a service) is equal or more than 3.

Calculate the condition that must be satisfied by  $\mu$ .

### Answer

$$\mu \geq 4$$

## **M/M/s( $\infty$ ) type**

### **Basic formulas**

**(1) Probability ( $P_0$ ) that the queuing system (waiting line and service station) is empty**

$$P_n = \frac{1}{\left[ \sum_{n=0}^{s-1} \frac{1}{n!} \rho^n \right] + \frac{1}{(s-1)!} \rho^s \frac{1}{s-\rho}}, \frac{\rho}{s} < 1$$

**(2) Probability ( $P_0$ ) that n units (persons, etc.) are in the queuing system (waiting line and service station)**

$$P_n = \frac{1}{n!} \rho^n P_0, n = 0, 1, 2, \dots, s-1$$

$$P_n = \frac{1}{s! s^{n-s}} \rho^n P_0, n \geq s$$

**(3) Mean number of units in the system (both waiting units and a unit receiving service)**

$$E(n) = \frac{\rho^{s+1}}{(s-1)!(s-\rho)^2} P_0 + \rho$$

**(4) Mean number of waiting units in the system**

$$E(m) = \frac{\rho^{s+1}}{(s-1)!(s-\rho)^2} P_0 = E(n) - \rho$$

**(5) Mean time spending in the system when a unit actually arrives at the system**

$$E(v) = \frac{1}{\mu} \frac{\rho^s}{(s-1)!(s-\rho)^2} P_0 + \frac{1}{\mu} = \frac{1}{\lambda} E(n)$$

**(6) Mean waiting time of an arrived unit**

$$E(w) = \frac{1}{\mu} \frac{\rho^s}{(s-1)!(s-\rho)^2} P_0 = \frac{1}{\lambda} E(m)$$

## Problem 4

There is a gas (petrol) station with two refueling equipments. Cars arrive at the gas station according to the Poisson distribution with mean arrival interval of 3 minutes. The distribution of service time obeys exponential distribution with the mean service time of 4 minutes.

- (1) Calculate the probability that there are no cars in the gas station.
- (2). Calculate the mean number of cars in the gas station.
- (3). Calculate the mean time from the arrival to the completion of service.

$$\text{Mean arrival rate } \lambda = \frac{1}{3} \text{ cars/min}$$

$$\text{Mean service rate } \mu = \frac{1}{4} \text{ cars/min}$$

$$\rho = \frac{\lambda}{\mu} = \frac{4}{3}, s = 2$$

$$(1) P_0 = \frac{1}{1 + \frac{4}{3} + \left(\frac{4}{3}\right)^2 \frac{1}{2 - \frac{4}{3}}} = \frac{1}{\frac{7}{3} + \frac{16 \times 3}{9 \times 2}} = \frac{1}{5} = 0.2$$

$$(2) E(n) = \frac{\left(\frac{4}{3}\right)^2}{1! \left(2 - \frac{4}{3}\right)^2} \times 0.2 + \frac{4}{3} = \frac{4^3 \times 3^2}{3^3 \times 2^2} \times \frac{1}{5} + \frac{4}{3} = 2.4$$

$$(3) E(v) = 4 \times \frac{\left(\frac{4}{3}\right)^2}{1! \times \left(2 - \frac{4}{3}\right)^2} \times 0.2 + 4 = 16 \times 0.2 + 4 = 7.2$$

## Problem 5

There are three medical doctors in clinic A. Assume that one working day consists of eight hours and medical doctors engage in their job for five days in a week. The medical treatment is conducted according to the principle of first in first served (FIFS). On average, a total of 24 patients visit this clinic every day. The distribution of medical treatment time obeyed exponential distribution with the mean medical treatment time of 40 minutes.

- (1) Calculate how many hours one doctor in this clinic engage in the job in a week.
- (2). Calculate the mean rate of operation for one doctor.
- (3). Calculate the patient's mean spending time in the hospital.

## Answer

$$(1) P_0 = \frac{1}{9}$$

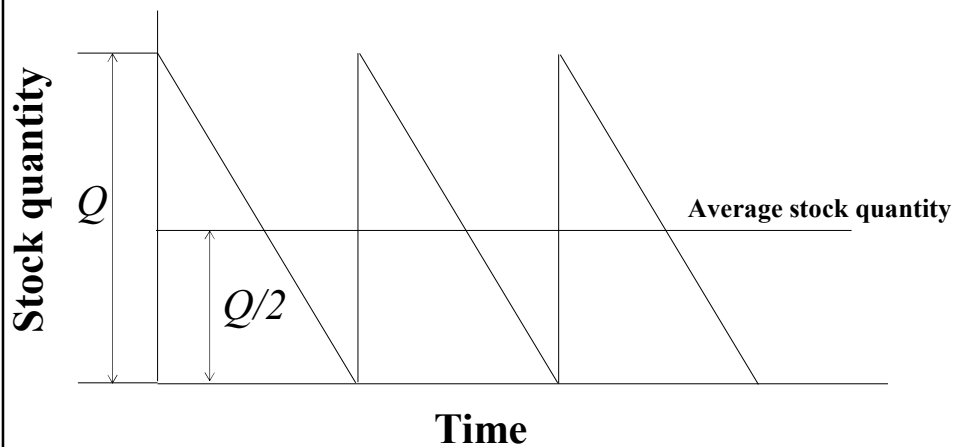
$$\therefore 40 \text{ h} \times (1 - 1/3) = 26.67 \text{ hours}$$

$$(2) \text{mean rate of operation for one doctor} = 2/3$$

$$(3) E(v) = \frac{26}{27} \text{ hours} = 60 \times \frac{26}{27} = 57.78 \text{ minutes}$$

## 6.Inventory management

### EOQ (Economical Order Quantity) model



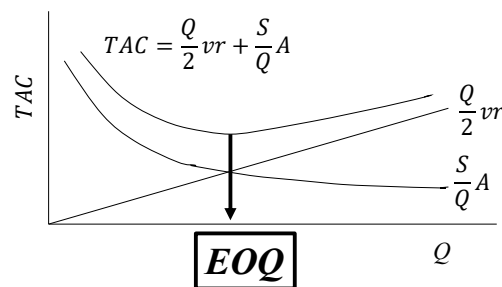
$Q$ : stock amount  
 $S$ : annual demand  
 $A$ : ordering cost per one order  
 $v$ : unit price (\$/unit)  
 $r$ : price for stock per \$  
 $vr$ : stock price for one unit  
 $TAC$ : total annual cost

$$TAC = \frac{Q}{2} vr + \frac{S}{Q} A \Rightarrow \text{minimization of } TAC$$

Obtain  $Q$  that minimizes  $TAC$

$$\frac{d(TAC)}{dQ} = \frac{1}{2} vr - \frac{SA}{Q^2} = 0$$

$$\frac{1}{2} vr = \frac{SA}{Q^2} \quad \therefore Q = \sqrt{\frac{2AS}{vr}} \quad \boxed{EOQ}$$



Reorder amount=(mean amount of demands during delivery period (lead time) and during order cycle) –order backlog –stock amount + safe stock amount

### Problem 1

Delivery period (lead time): 2 months  
 Order cycle: 1 month  
 Order backlog: 800 units  
 Stock amount: 85 units  
 Safe stock amount;120 units  
 Mean demand: 600 units/month  
 Calculate reorder amount.

### Problem 2

$S$ :demand per 30 days is 1500 units of products.  
 $vr$ : cost for stock of one unit of product per day is \$25.  
 $A$ : ordering cost per one order is \$500.  
 Obtain EOQ.



**Answer**

|                             |
|-----------------------------|
| Reorder amount = 1035 units |
|-----------------------------|

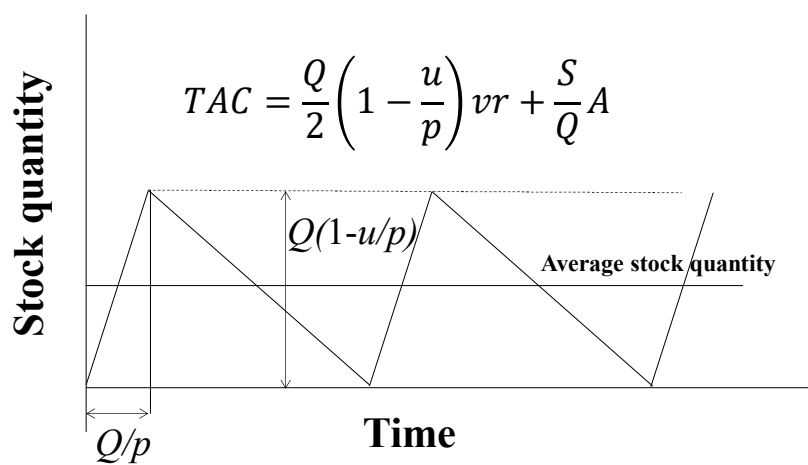
**Answer**

|                      |
|----------------------|
| $Q = 44.72 \cong 45$ |
|----------------------|

 $Q$ : stock amount $S$ : annual demand $A$ : ordering cost per one order $v$ : unit price (\$/unit) $r$ : price for stock per unit price (\$) $vr$ : stock price for one unit $TAC$ : total annual cost $p$ : quantity of partition delivery per unit term $u$ : quantity of consumption per unit term

$$TAC = \frac{Q}{2} \left( 1 - \frac{u}{p} \right) vr + \frac{S}{Q} A \Rightarrow \text{minimization of } TAC$$

Obtain  $Q$  that minimizes  $TAC$



$$\text{Average stock quantity} = \frac{1}{2} Q(1-u/p)$$

$$\frac{d(TAC)}{d Q} = \frac{1}{2} \left( 1 - \frac{u}{p} \right) vr - \frac{SA}{Q^2} = 0$$

$$\frac{1}{2} \left( 1 - \frac{u}{p} \right) vr = \frac{SA}{Q^2}$$

$$\therefore Q = \sqrt{\frac{2AS}{\left( 1 - \frac{u}{p} \right) vr}}$$

### Problem 3

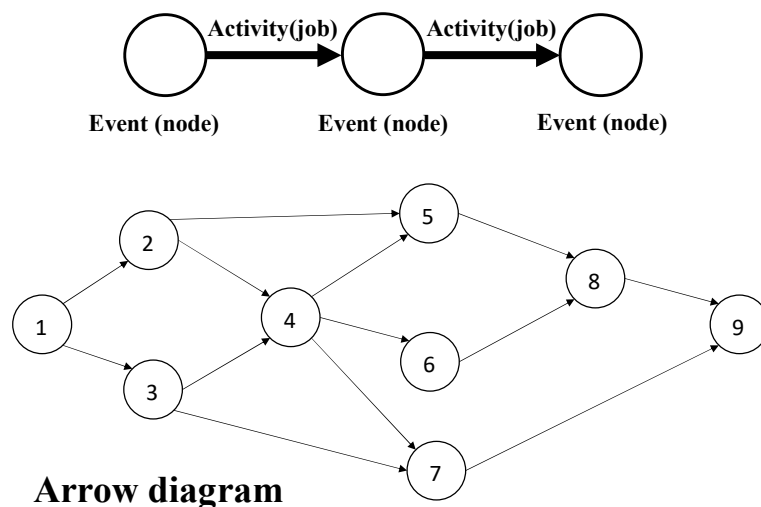
One manufacturer delivers 20000 parts to a automotive manufacturer. This company can manufacture 30000 parts if it takes a special means. The cost of annual stock per one part is \$5. The cost of arrangement (preparation) for order is \$3000/one arrangement (preparation). Operation days per one year is assumed to be 300 days. Obtain optimum manufacturing lot  $Q$  and manufacturing period  $T$ .

### Answer

$$Q = 6 \times 2.45 \times 10^4 = 147000 \text{ parts}$$
$$T = 7.35 \text{ days}$$

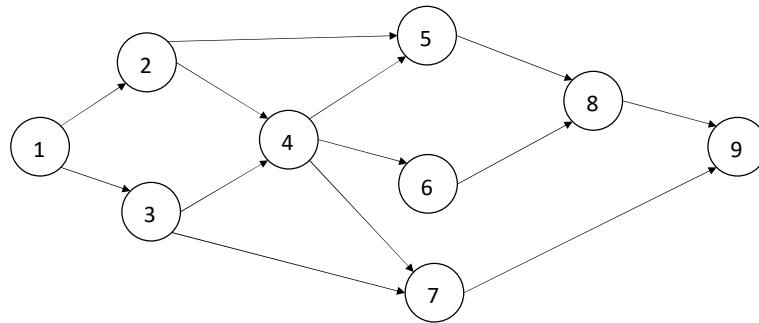
## 7.PERT • CPM

### Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM)

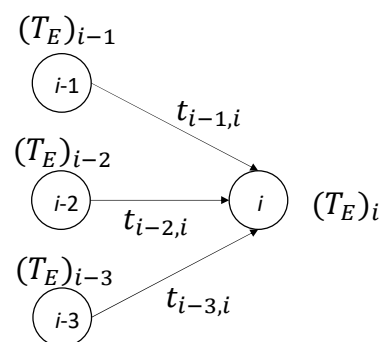


## Problem 1

Obtain the preceding event and the following events of event 4.

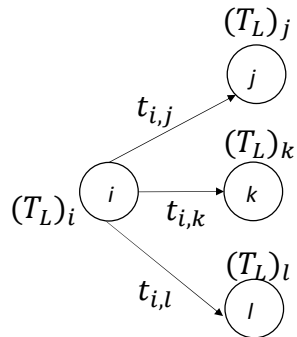


## Earliest node time



$$(T_E)_i = \max((T_E)_{i-1} + t_{i-1,i}, (T_E)_{i-2} + t_{i-2,i}, (T_E)_{i-3} + t_{i-3,i})$$

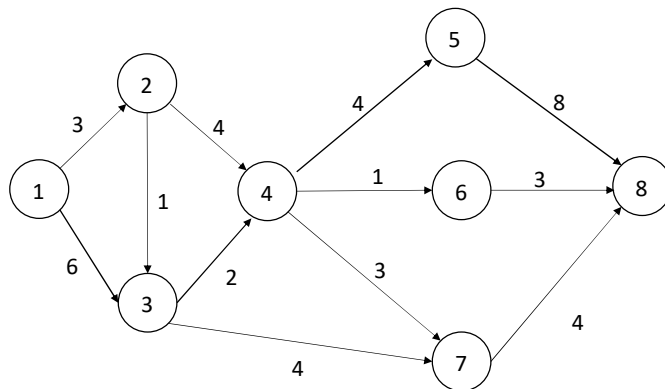
### Latest node time

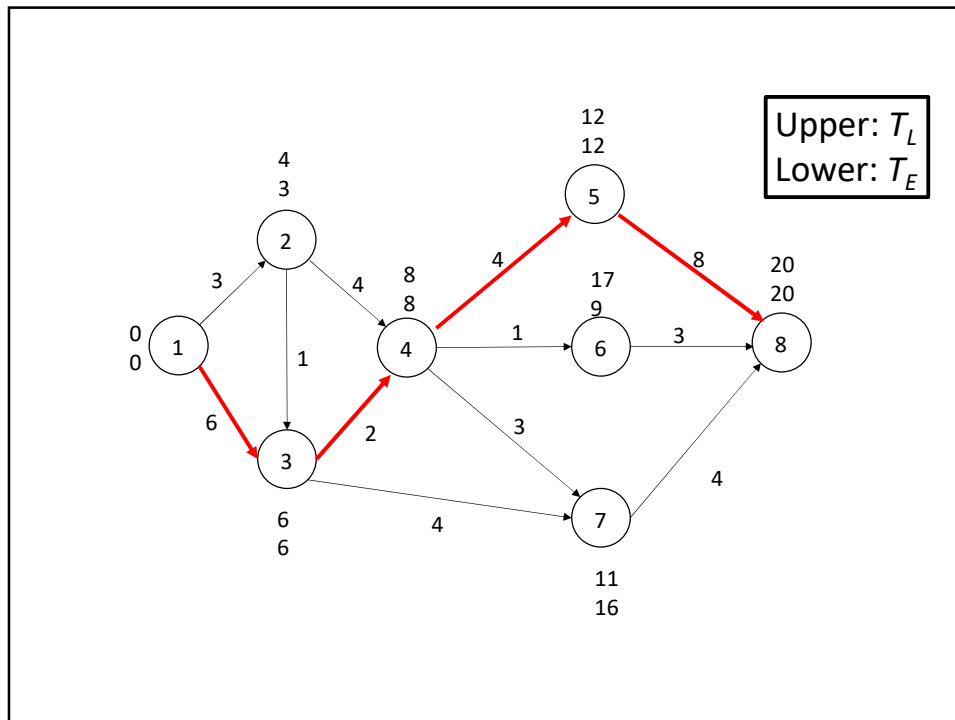


$$(T_L)_i = \min((T_L)_j - t_{ij}, (T_L)_k - t_{ik}, (T_L)_l - t_{il})$$

### Problem 2

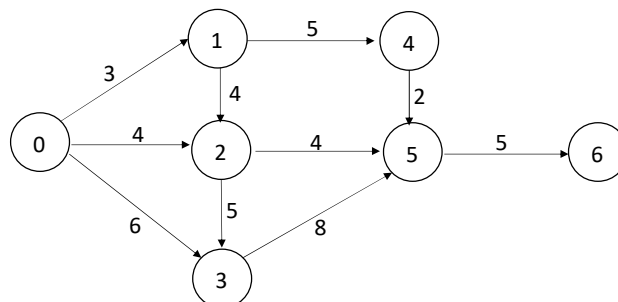
Obtain the CP of the following arrow diagram.

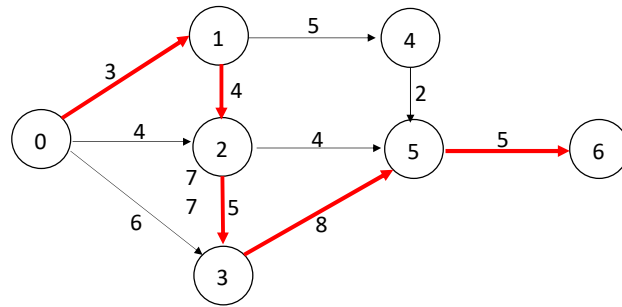




### Problem 3

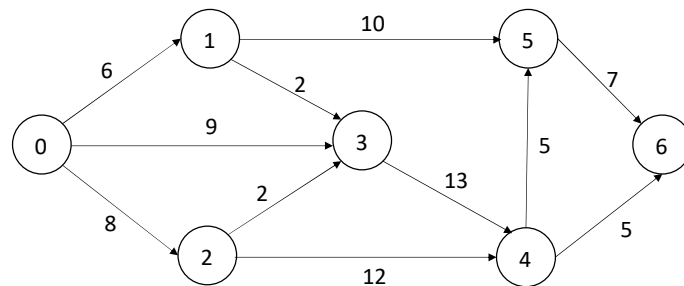
Obtain the CP of the following arrow diagram.



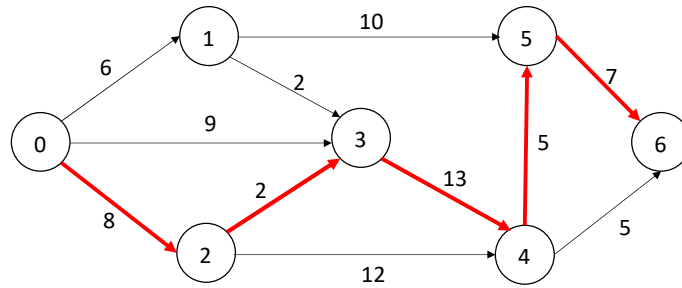


## Problem 4

Obtain the CP of the following arrow diagram.

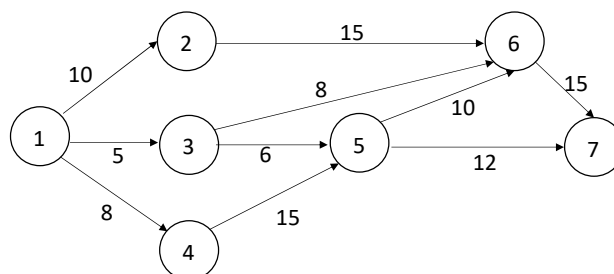


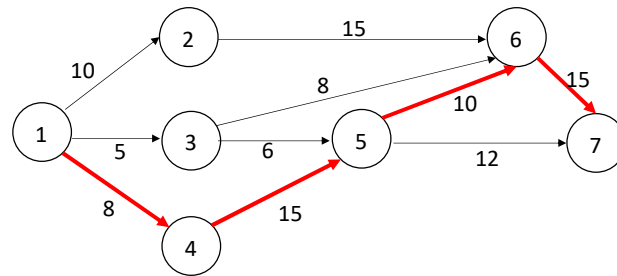




## Problem 5

Obtain the CP of the following arrow diagram.





## Problem 6

There is a project network as shown in the table below. In this table, optimistic value:  $a$ , most probable value:  $m$ , and pessimistic value:  $b$  of each activity is shown. Draw a arrow diagram of this project network, and obtain CP (critical path). Assume that the expected completion of event 9 is  $T_s=83$ , and calculate the probability that the project is completed until the expected completion  $T_s$ .

| preceding event | following event | $a$ | $m$ | $b$ |
|-----------------|-----------------|-----|-----|-----|
| 1               | 2               | 3   | 8   | 10  |
| 1               | 3               | 5   | 10  | 15  |
| 2               | 4               | 6   | 10  | 24  |
| 3               | 2               | 2   | 5   | 10  |
| 3               | 5               | 4   | 16  | 20  |
| 3               | 6               | 6   | 15  | 18  |
| 4               | 5               | 8   | 14  | 24  |
| 4               | 9               | 10  | 20  | 30  |
| 5               | 7               | 4   | 10  | 20  |
| 5               | 9               | 6   | 10  | 15  |
| 6               | 7               | 2   | 4   | 10  |
| 7               | 8               | 5   | 8   | 15  |
| 8               | 9               | 7   | 10  | 20  |

### Prediction of completion time of a project

Completion time of each activity (job)

optimistic value:  $a$ , most probable value:  $m$ ,

pessimistic value:  $b$

$$\text{Mean } E(t) = \frac{a + 4m + b}{6}$$

$$\text{Variance } V(t) = \left( \frac{b - a}{6} \right)^2$$

$T_E$ : estimated value of task completion time

$\sigma_{T_E}^2$ : estimation error of  $T_E$

$T_S$ : predetermined completion time

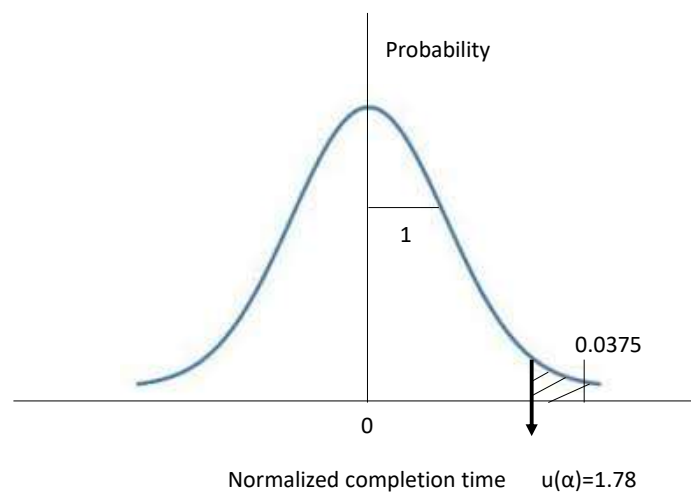
$P$ : probability that a project is completed

within  $T_S$  ( $t \leq T_S$ ) =  $\alpha$

$$u = \frac{t - T_E}{\sqrt{\sigma_{T_E}^2}}$$

$\Rightarrow$

$\alpha$  is obtained from normal distribution table.

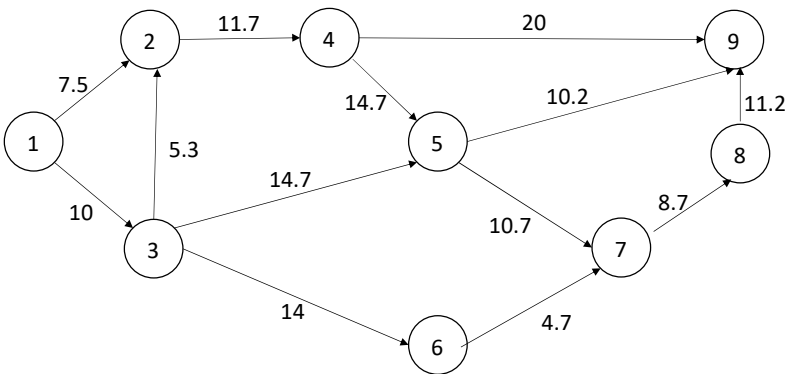


$$u = \frac{t - T_E}{\sqrt{\sigma_{T_E}^2}} = \frac{T_S - 72.3}{6} = \frac{83 - 72.3}{6} = 1.78$$

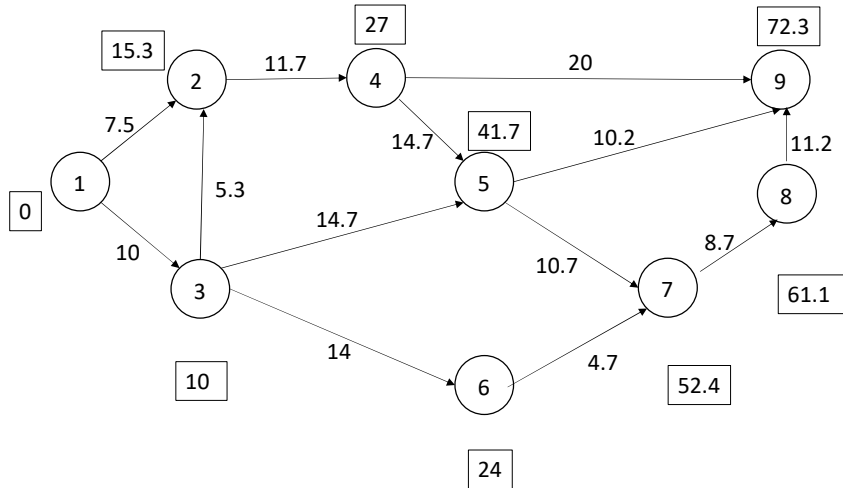
Calculation of  $T_E$  and  $\sigma^2$

| preceding event | following event | $a$ | $m$ | $b$ | $T_E$ | $\sigma^2$ |
|-----------------|-----------------|-----|-----|-----|-------|------------|
| 1               | 2               | 3   | 8   | 10  | 7.5   | 1.4        |
| 1               | 3               | 5   | 10  | 15  | 10.0  | 2.8        |
| 2               | 4               | 6   | 10  | 24  | 11.7  | 9.0        |
| 3               | 2               | 2   | 5   | 10  | 5.3   | 1.8        |
| 3               | 5               | 4   | 16  | 20  | 14.7  | 7.1        |
| 3               | 6               | 6   | 15  | 18  | 14.0  | 4.0        |
| 4               | 5               | 8   | 14  | 24  | 14.7  | 7.1        |
| 4               | 9               | 10  | 20  | 30  | 20.0  | 11.1       |
| 5               | 7               | 4   | 10  | 20  | 10.7  | 7.1        |
| 5               | 9               | 6   | 10  | 15  | 10.2  | 2.3        |
| 6               | 7               | 2   | 4   | 10  | 4.7   | 1.8        |
| 7               | 8               | 5   | 8   | 15  | 8.7   | 2.8        |
| 8               | 9               | 7   | 10  | 20  | 11.2  | 4.7        |

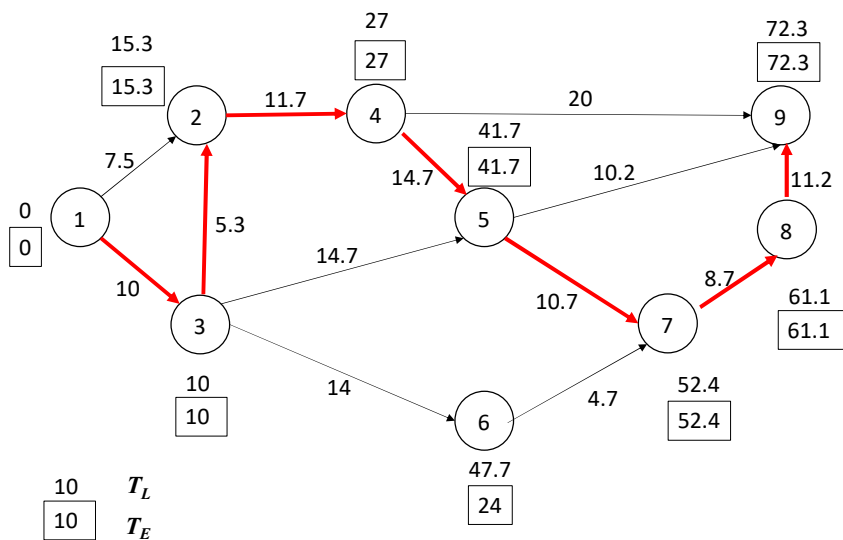
Draw of a project network (arrow diagram)

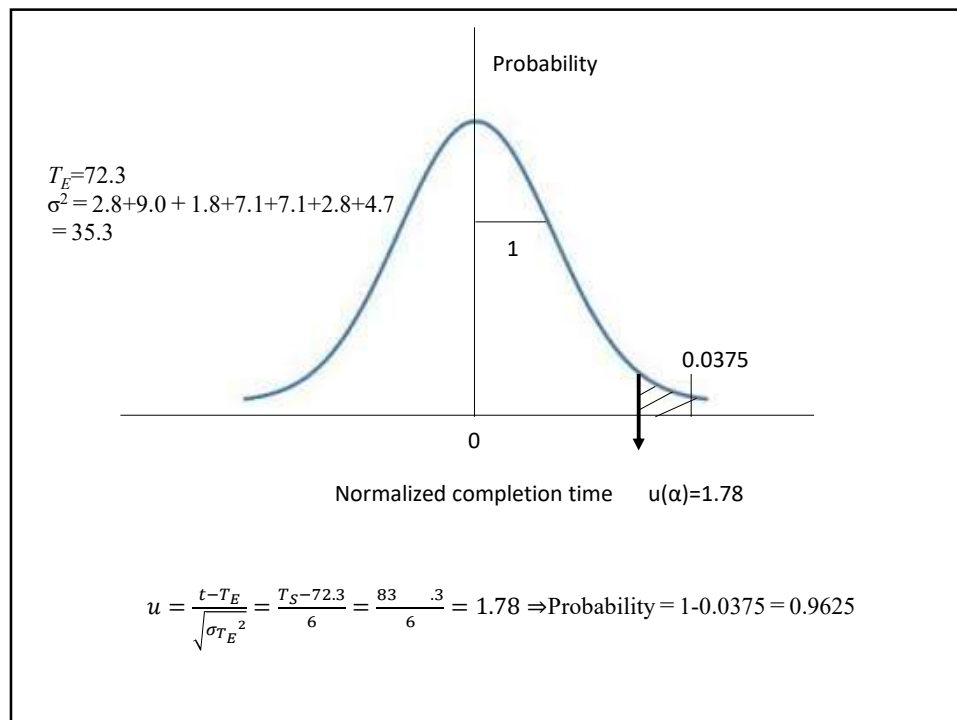


### Calculation of $T_E$



### Calculation of $T_L$ and Obtain CP





## Recommended reference

**Levent Kandiller: Principles of Mathematics in Operations Research (International Series in Operations Research & Management Science), Springer, 2010**

**Project**

Describe how operational research is effective for company management

Chap1: Problems 2 and 3

Chap 2: Problem 2

Chap 3: Problem 2

Chap 4: Problem 2

Chap 5: Problems 2, 3, and 5

Chap 6: Problem 3

Chap 7: Problems 3-5

*Hand written A4 paper*

**Deadline of Report**

6, February, 2025

From 11:00 to 12:00 (will not be accepted at pther than this term)

RA244